

Andrew Olvera

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RESEARCH INTERESTS

Agile, generalist control policies; reinforcement learning; generative world models. Current focus: scalable pretraining for control, using privileged classical controllers as teachers to generate synthetic supervision and distilling it into general policies.

EDUCATION

York College of Pennsylvania

Aug 2023 – May 2026

B.S. in Computer Science, York, PA

- Coursework: Data Structures, Algorithms, Linear Algebra, Calculus, Probability & Statistics, Computer Vision, Computer Graphics

Cabrini University

Aug 2021 – May 2023

Undergraduate coursework, Radnor, PA; transferred to York College

RESEARCH EXPERIENCE

Starscream: Supervised Pretraining for Route-Conditioned Agile Flight

2026 – Present

Independent research

- Pretrained a 5.84M-parameter causal transformer policy using MPCC expert labels, DAgger, DART, and an auxiliary task-delta prediction head.
- Conditioned the policy on recent state and control history, six ordered gates, privileged vehicle settings, and a test-time speed command.
- A selected checkpoint completed 94.3% of 1,200 simulated starts across 100 courses; 68.8% were clean and timely (one training seed; real100-v2 development benchmark).
- Comparing history MLP, LSTM, and transformer policies across independent training seeds, with separate measures of completion, pace, and missed-gate recovery.

Tells: Opponent Beliefs from Play History in a MicroRTS World Action Model

2026 – Present

Independent research

- Built action-conditioned latent world models and state/action tokenizers for MicroRTS, with complete-state and fog-visible representations.
- Distilled a 16-code opponent belief from fog-of-war history using privileged training targets; evaluated predictive utility separately from policy outcomes.
- Trained PMPO actor-critic policies with eight-step imagined rollouts in a frozen learned world, alongside real-engine data collection; compared belief-conditioned and separately trained no-belief policies.

Learning Dynamics in RLVR Near Capacity

2025 – 2026

Independent research

- Studied RLVR on Qwen3 0.6B (GSM8K), comparing GRPO and DAPO across group sizes, then binary, structured, and rank-based rewards on a fixed baseline.
- Exploratory single-seed runs reached roughly 69–72% accuracy from a 68% SFT baseline; tracked reward variation as binary rewards saturated.
- Analyzed token-level advantages, dynamic clipping, and reward structure to study when the optimizer retains a useful learning signal.

PREPRINTS IN PREPARATION

A. Olvera. *Starscream: Supervised Pretraining for Route-Conditioned Agile Flight*. In preparation, 2026.

A. Olvera. *Tells: Opponent Beliefs from Play History in a MicroRTS World Action Model*. In preparation, 2026.

A. Olvera. *Learning dynamics in RLVR near capacity*. In preparation, 2026.

AWARDS, TALKS & SERVICE

- Dean's List, York College of Pennsylvania · Fall 2025, Spring 2026

INDUSTRY EXPERIENCE

Sapio Sciences

May 2025 – Present

Application Engineer Intern, York, PA

Dataforma

May 2025 – Jan 2026

Software Engineer Intern, Web Development, York, PA

- Refactored legacy JSP modules into a Vue.js frontend and Kotlin backend, improving maintainability and enabling new feature development.
- Partnered with QA to debug defects, execute test plans, and validate deployments.

TECHNICAL SKILLS

ML & RL: PyTorch, Hugging Face Transformers, TRL, vLLM; imitation learning (DAgger, DART), policy gradients (PPO, GRPO, DAPO), RLVR, RLHF/DPO, model-based RL, self-supervised learning, computer vision

Control & Simulation: Control theory, MPC/MPCC (acados), ROS, Flightmare, IsaacSim, CARLA, Basilisk, MATLAB/Simulink

Languages: Python, C++, C, Java, Kotlin, JavaScript, C#, SQL

Tools: Linux, Git, Docker, Bash, HDF5, WebDataset